Edgar J. Andrade-Lotero, Ph.D.

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Professional Summary

Researcher and academic with over two decades of experience at the intersection of logic, cognitive science, and artificial intelligence. Holds a Ph.D. in Logic from the University of Amsterdam and has published in leading venues including Philosophical Transactions of the Royal Society B, PLOS ONE, and Topics in Cognitive Science. Current research spans collective intelligence, NLP with large language models, and reinforcement learning. Experienced in supervising graduate students and teaching across mathematics, AI, and philosophy of language.

Experience

Associate Professor 2019 – Present
Universidad del Rosario *School of Sciences and Engineering*

Postgraduate Researcher 2025 – Present
University of California, Davis *Department of Communication*

Expert Consultant 2024 – 2025
Turing.com

Associate Professor 2010 – 2019
Universidad del Rosario *Faculty of Natural Sciences and Mathematics*

Auxiliary Professor July 2006 – 2010
Universidad del Rosario *School of Social Sciences*

Lecturer February 2006 – June 2006
Universidad Sergio Arboleda *School of Mathematics*

Lecturer February 2004 – June 2004
Universidad Sergio Arboleda *School of Mathematics*

Lecturer 2003 – 2004
Universidad Nacional de Colombia *Department of Mathematics*

Education

University of Amsterdam
Ph.D. in Logic. Thesis: Models of Language: Towards a practice-based account of information in natural language. *2012*

University of Amsterdam
M.Sc. in Logic. Thesis: Meaning and Form in Event Calculus. *2006*

Universidad Nacional de Colombia
B.A. in Mathematics. *2002*

Publications: Research Articles

- E. J. Andrade-Lotero, C. Alvarez, and G. Obando, "A paradox in Brian Arthur's solution to his El Farol Bar problem," Advances in Complex Systems, vol. 27, no. 8, p. 2550014, 2025.
- J. Piza-Londono and E. J. Andrade-Lotero, "Leveraging semantic parsing using text embeddings and reinforcement learning," 2024 IEEE Latin American Conference on Computational Intelligence (LA-CCI), pp. 1-6, 2024.
- R. L. Goldstone, E. J. Andrade-Lotero, R. Hawkins, and M. Roberts, "The emergence of specialized roles within groups," Topics in Cognitive Science, vol. 16, pp. 257-281, 2023.
- E. J. Andrade-Lotero, J. A. Velasco, J. Ortiz-Duque, and R. L. Goldstone, "The division of linguistic labor for offloading conceptual understanding," Philosophical Transactions of the Royal Society B, vol. 378, no. 20210360, 2022.
- E. J. Andrade-Lotero and R. L. Goldstone, "Self-organized division of cognitive labor," PLOS ONE, vol. 16, no. 7, 2021.
- E. J. Andrade-Lotero and J. Ortiz-Duque, "A mis-conception of logic in its diffusion across non-mathematical environments," Linguistic and Philosophical Investigations, vol. 18, pp. 77-96, 2019.
- C. Alvarez, E. J. Andrade-Lotero, and V. Gauthier, "Modelos epidemiologicos en redes: Una presentacion introductoria," Boletin de Matematicas, vol. 22, no. 1, pp. 21-37, 2015.
- E. J. Andrade-Lotero, "Una gramatica logica para explicar algunos aspectos de la semantica del espanol," Boletin de Matematicas, vol. 20, no. 1, pp. 21-50, 2013.
- E. J. Andrade-Lotero and C. Dutilh, "A contentious trinity: Levels of entailment in Brandom's pragmatic inferentialism," Philosophia, vol. 40, no. 1, pp. 41-53, 2012.
- E. J. Andrade-Lotero and C. Dutilh, "Validity, the squeezing argument and alternative semantic systems: The case of Aristotelian syllogistic," Journal of Philosophical Logic, vol. 41, no. 2, pp. 387-418, 2012.

- E. J. Andrade-Lotero and E. Becerra, "Establishing connections between Aristotle's natural deduction and first-order logic," History and Philosophy of Logic, vol. 29, no. 4, pp. 309-325, 2008.
- E. J. Andrade-Lotero and E. Becerra, "Corcoran's Aristotelian syllogistic as a subsystem of first-order logic," Revista Colombiana de Matematicas, vol. 41, no. 1, pp. 67-80, 2007.

PUBLICATIONS: BOOKS AND TRANSLATIONS

- L. Gamut, Logica, lenguaje y significado: logica intensional y gramatica logica, E. Andrade-Lotero and C. M. (translators), Eds. Bogota: Editorial Universidad del Rosario, 2010.
- E. J. Andrade-Lotero, P. Cubides, C. Marquez, E. J. Vargas, and D. Cancino, Logica y Pensamiento Formal. Editorial Universidad del Rosario, 2008.

PUBLICATIONS: CONFERENCE PROCEEDINGS (EDITED)

- M. S. Dora Achouroti, Edgar J. Andrade-Lotero, Eds., Proceedings of the Amsterdam Graduate Philosophy Conference: Meaning and Truth, X-2010-01, University of Amsterdam, ILLC Publications, 2010.
- M. S. Dora Achouroti, Edgar J. Andrade-Lotero, Eds., Proceedings of the Graduate Philosophy Conference on Normativity, X-2009-02, University of Amsterdam, ILLC Publications, 2009.

PUBLICATIONS: CONFERENCE PAPERS

- E. J. Andrade-Lotero and R. L. Goldstone, "An empirical study of the El Farol Bar problem," Collective Intelligence, 2024.
- E. J. Andrade-Lotero and R. L. Goldstone, "Self-organized division of cognitive labor," Proceedings of the 41st Annual Conference of the Cognitive Science Society, 2019, pp. 91-97.
- E. J. Andrade-Lotero and R. L. Goldstone, "Self-organized division of cognitive labor," Collective Intelligence, 2019.
- E. J. Andrade-Lotero and J. Ortiz, "The proper approach to the division of linguistic labor," Sixth International Conference on Philosophy of Language and Linguistics, 2019.
- E. J. Andrade-Lotero, "Translations and structures: Some reflexions about teaching the semantics of first-order logic," Philogica IV, 2016.
- E. J. Andrade-Lotero, "Explicating compositionality," Philogica III, 2014.
- E. J. Andrade-Lotero, "On truth conditions that depend on performatives," Third International Conference on Philosophy of Language and Linguistics, 2013.
- E. J. Andrade-Lotero, "El principio de composicionalidad y la competencia linguistica," IV Congreso Colombiano de Filosofia, 2012.
- E. J. Andrade-Lotero, "Compositionality from a 'use-theoretic' perspective," 32nd International Wittgenstein Symposium, 2009.
- E. J. Andrade-Lotero and C. Dutilh, "A reconstruction of the hierarchical relations between incompatibility entailment, committive and permissive consequence," Proceedings of the Workshop on Bob Brandom's Recent Philosophy of Language: Towards an Analytic Pragmatism, 2009, pp. 90-98.

STUDENTS SUPERVISED

- E. J. Lopez-Lopez and S. Gaitan-Rubio, Modelacion de precios por medio del juego de las minorias, Masters in Applied Mathematics and Computer Science, 2024.
- E. Munoz, Gestion activa de portafolios de cripto activos utilizando tecnicas de aprendizaje por refuerzo, Masters in Applied Mathematics and Computer Science, 2024.
- J. Piza-Londono, Aprendizaje por refuerzo de un parser semantico optimo en DRT, Masters in Applied Mathematics and Computer Science, 2024.
- J. V. Tamayo-Ramirez, G. A. N. Carcamo, and R. D. Rico-Gonzalez, Automatizacion de procesos de cualificacion en IQAP mediante chatbot de IA para ingreso a universidades internacionales, Masters in Applied Mathematics and Computer Science, 2024.
- J. A. Velasco, Hacia una cartografia del significado: Una exploracion por los roles del sentido en el Tractatus-Logico-Philosophicus, Bachelors in Philosophy, 2019.
- M. P. Gomez, La incompletitud de los fundamentos de las matematicas, Bachelors in Philosophy, 2018.
- M. G. Algarra, Inconmensurabilidad e interpretacion: Una propuesta de comparacion entre teorias cientificas, Bachelors in Philosophy, 2007.

COURSES TAUGHT

- 2017-Present: NLP with Large Language Models, Reinforcement Learning, Artificial Intelligence, Logic for Computer Science, Theory of Computation, Set Theory, Linear Algebra -- Universidad del Rosario.
- 2007-2016: Differential Calculus, Integral Calculus, Probability, Statistics, Philosophical Logic, Philosophy of Language -- Universidad del Rosario.
- 2006: History of Mathematics, Theory of Computation -- Universidad Sergio Arboleda.
- 2003: Mathematical Logic -- Universidad Sergio Arboleda.

- 2002: Logic for Psychologists -- Universidad Nacional de Colombia.

CERTIFICATIONS

- Generative AI Engineering, Coursera
- Generative AI with Large Language Models, Coursera
- Machine Learning Engineering for Production, Coursera
- Deep Learning, Coursera
- Reinforcement Learning for Trading, Cognitive Class
- Machine Learning with Python, Coursera
- Python for Data Science, AI & Development, Coursera

SKILLS

Programming Languages: Python, R, JavaScript, LaTeX

Spoken Languages: English (C1, TOEFL), French (A4, DELF)

AWARDS & HONORS

- 2022: Visiting Scholar Grant, Universidad del Rosario
- 2020: Research Seedbed Grant, Universidad del Rosario
- 2018: Visiting Scholar Grant, Universidad del Rosario
- 2018: Seed Capital Grant, Universidad del Rosario
- 2004: Huygens Scholarship for Outstanding Foreign Students, Netherlands Organization for International Cooperation in Higher Education